MET253: Fortran For Scientific Computing

Instructor: Dr. rer. nat. Edmund Ilimoan Yamba

Kwame Nkrumah University of Science and Technology
Department of Meteorology and Climate Science

email: eiyamba@knust.edu.gh

"If you want to fly, give up everything that weighs you down"

FORTRAN

FORMULA TRANSLATION

Course Outline

1. FORTRAN BASICS
   - Background to Fotran Programming
   - Structure of a Fortran Program
   - Creating, Compiling and Running a fortran code
   - Features of Fortran Program
   - Operators
   - Intrinsic Fortran Functions
   - Formatting Fortran Output
1. Control Structures
   - IF Statements, IF Constructs
   - Do loops
   - CASE Constructs
2. Arrays
3. Reading and Writing Files
4. Functions
5. Subroutines
6. Modules
7. Visualizing Data Using gnuplot

Course Objectives

1. Main Objective is that, students will be:
   - Familiar with Fortran programming and be able to use it for **meteorological applications**
2. Specific objectives are that students will be to:
   - Understand the fundamentals of Fortran programming
   - Read files and write into files in FORTRAN
   - Use control structures, functions, subroutines and modules in fortran code
   - Write a well-structured Fortran program
   - Compile, run and debug Fortran codes
   - Visualize FORTRAN outputs with gnuplot.

Learning outcome

By the end of this course, students will be able to:

i. Discover the power of Fortran for scientific computing
ii. Learn how to use Fortran to solve numerical problems
iii. Explore data types and scientific computing fortran libraries
iv. Code in Fortran according to best practices
v. Evaluate the performance of a fortran code
vi. Identify and select the most relevant data structures for fortran computing

Reading list

1. Introduction to Programming with Fortran
   By: Ian Chivers
2. Fortran 95/2003 for Scientists & Engineers
   By: Stephen Chapman
3. Modern Fortran Explained (Numerical Mathematics and Scientific Computation)
   By: Michael Metcalf and John Reid
4. Modern Fortran Explained
   By: Michael Metcalf
5. Modern Fortran in Practice
   By: Arjen Markus
6. https://www.fortrantutorial.com/basics/index.php
7. https://www.youtube.com/playlist?list=PLNmACoI6lYY5_S8I4K29V2oI95nOCzQU8

Reading list

- Tutorials
  - https://math.hawaii.edu/wordpress/fortran-lessons/
  - https://www.tutorialspoint.com/fortran/
  - You can ask questions here: https://stackoverflow.com/ for answers.
- Online Books
  - Fortran 90/95 Explained 2nd Edition, by by Michael Metcalf and John K. Reid
  - Modern Fortran Explained (Numerical Mathematics and Scientific Computation) 4th Edition by Michael Metcalf, John Reid and Malcolm Cohen.

Notice to students on the lecture notes

- The lecture Notes provided here are additional resource for students and is not guaranteed to be complete, or up-to-date; and should not be viewed as a replacement for taking notes in class.
- Should the notes conflict with anything I state or demonstrate in class, then what is presented in class takes precedence.
- Please bring any discrepancies to my attention.

Background: A computer program

A computer program is a set of instructions for the computer to execute sequentially. The particular set of rules for coding the instructions to a computer is called a programming language.

Steps involved:

- Initially the program source code is written in plain text by the programmer using text editor. Fortran 90 files typically have extension (.f90).
- This file is then compiled (by existing application, called the compiler) to translate the plain text of the source code to binary code for the processor. The executable files have extension (.exe).
- The binary executable file is then run to get the output/results in plain text.

Background:Fortran Language

Fortran (FORmula TRANslator) was the first high-level programming language proposed in 1953 by John Backus and the first program was run in April 1957. Fortran is use mainly within the scientific community for solving problems that has a significant arithmetic content.

Versions/Standards

- Fortran 66 - first standard
- Fortran 77 - character handling
- Fortran 90 - array and modular programming
- Fortran 95 - functional programming
- Fortran 2003 - object-oriented and generic programming
- Fortran 2008 - concurrent programming.

Fortran Program Structure

The structure of a FORTRAN program has the following elements:

[PROGRAM name]

[IMPLICIT NONE]

[Declaration-part]

[execution-part]

[subprogram-part]

END [PROGRAM name]

Everything in square brackets [ ] is optional. However, it is good programming practice to put the name of the program in both header and END statements.

T

he program name must :

- Begin with a letter (upper/lower case)
- It contains letters. digits 0-9, and underscore characters
- Maximum of 31 characters.

Example program:Product of two variables

```
PROGRAM firstprogram                                  ! start of program
!
  !This program finds product of two real variables
!
  IMPLICIT NONE                  ! All variables used must be declared
!
  REAL :: x,y,z                                       ! Declare variables
! variable assignment
  x = 5.1                                             ! Assign x
  y = -17.2                                           ! Assign y
!execution
  z = x*y                                             ! Assign (x*y) to z
  PRINT*,'The product of x and y is', z          ! Print out z
!
END PROGRAM firstprogram                              ! end of program
```

Creating a fortran program

- Either open a pre-existing source code or create it and save using text editor and should have a name that ends in .f90.
- A text editor is a type of program used for editing plain text files.
- Examples of some text editors are (emacs, gvim, vi, gedit, kile, xemacs etc)
- On command line and inside your program directory, open the file by typing:
- emacs prog1.f90 where prog1 is the filename and emacs is the text editor
- Emacs shows blank page since prog1 is a new file.

Compiling fortran program

- The role of fortran compilers is to translate the plain text of the source code to binary code for the processor.
- Some free available compilers are (g95, gfortran, ifort, etc)
- During compilation the program is checked for syntax errors and, if none are found, an executable file is created
- The default name for the executable file will be a.out but you can also specify a name with different extension.
- In the same directory containing the program file type of one these commands:
  1. g95 -o prog1.exe prog1.f90 this produces executable file called prog1.exe
  2. g95 prog1.f90 this produces executable file called a.out

Running fortran program

- If compilation step is carried out successfully, then the executable prog1.exe or a.out will appear when the files are listed with ls.
- You can then run the program with the command ./prog1.exe or ./a.out
- The ./ before the file name tells the computer that the executable file is in the current directory.

Example program: Radius and perimeter of cycle

```
PROGRAM maths                                   ! start of program
  !This program finds the radius and perimeter of a cycle
  IMPLICIT NONE                    ! All variables used must be declared
  REAL, PARAMETER :: pi = 3.143          ! parameter declaration
  REAL :: rad, area, peri                  ! Declare variables to be used
  PRINT*, 'Enter the radius of the cycle'
  READ*, rad
  area = pi*rad**2                                  ! assign area
  peri = 2*pi*rad                                   ! assign peri
  PRINT*,'The area of a cycle of radius', rad 'is', area
  PRINT*,                                            ! print blank line
  PRINT*,'The perimeter of a cycle of radius', rad 'is', peri
END PROGRAM maths                                    ! end of program
```

Features of Fortran Program 1/8

IMPLICIT NONE

This is very important non-executable fortran statement used to disable the default typing provisions of Fortran. When this statement is used in the program, any variable that does not appear in an explicit type declaration statement is considered error. If omitted, fortran uses implicit declaration where a variable is considered integer if the first letter of its names lies in the range of I - N and will be declared to be a real variable otherwise.

IMPLICIT NONE should appear after the Program name and before any type declaration statement.

IMPLICIT NONE will be use in this course to write safe programs.

Features of Fortran Program 2/8

STATEMENT

- Fortran statement layout is one-per-line separated with blank lines for clarity.
- There may be more than one statement per line, separated by a semicolon(;) e.g
  A = 1; B = 10; C = 5
- Line may be up to 132 characters long.
- If the statement is too long to fit on one line, the continuation character (&, ampersand) is used as last character and then statement continue on next line; e.g
  Rad = Deg * 3.14 &
  /180.0
  The above is equivalent to Rad = Deg * 3.14/180.0

Features of Fortran Program 3/8

Fortran Data type

- A Fortran 90 data type may be an integer, real, logical, complex, and character string
- An integer data type is a string of digits with an optional sign: 12, -20, 0, +2333
- A real data type has two forms decimal and exponential

  The decimal form consist of string of digits with exactly one decimal point. e.g 1.4, -13.5, +0.04, 33333.5

  In the exponential form, start with an integer/real. followed by a E/e, followed by an integer (i.e., the exponent). e.g 15E3 ($12x10^3$), equivalent to 15e3 ($15x10^3$)

Features of Fortran Program 4/8

Fortran Data type cont.

- A logical data type is either .TRUE. or .FALSE.
  Note that the periods surrounding TRUE and FALSE are required!
- A complex data type consists of two numeric constants separated by comma and enclosed in parentheses. e.g (1.,0.) equivalent to 1 + 0i and (0.7071,0.7071) equivalent to 0.7071 + 0.7071i
- A character/string data type is a string of characters enclosed between two double or two single quotes. e.g "abc" produces abc
  The length of a character string is the number of characters between the quotes including blank space.

Features of Fortran Program 5/8

Variables

- A variable is an entity that is used to store a value, comparable to the use of variable in algebra.
- Must be between 1 and 31 characters in length. The first character must be a letter and the remaining if any, may be letters, digits or underscore.
- The variables are case insensitive and therefore NAME and naMe are the same.
- Fortran 90 does not have reserved words and therefore Fortran keywords can be used as variable names but avoid it.
- The type of a variable is declared in the declaration statement.
- Variable can be given a value during the execution of the program - by assignment or by a READ statement.

Features of Fortran Program 5/8

Valid Variables names

x
my_variable
Results17
STUDENT

Invalid Variables names

1x first character is a number
my$variable '$' is not letter, numeral or underscore
my_really_really_long_variable_name more than 31 characters

Features of Fortran Program 6/8

Declaration of Variables

- Fortran 90 uses the following for variable declaration, where type-specifier is one of the fortran following keywords: INTEGER, REAL, LOGICAL, COMPLEX and CHARACTER, and list is a sequence of identifier separated by commas.

  type-specifier :: list

  Example:
  INTEGER :: x, total
  REAL :: average, z_mean
- CHARACTER variables require the string length
- The keyword CHARACTER must be followed by the length attribute (LEN = integer_value), where integer_value is the length of the string.

Features of Fortran Program 7/8

Declaration of Variables cont

- If the length of the string is 1 then the CHARACTER can be declared without the LEN attribute.
  EXAMPLE:
  - CHARACTER (LEN = 20) :: answer, name
    Variables answer and name can hold up to 20 characters.
  - CHARACTER :: phy
    Variable phy can hold only ONE character (i.e length is 1)
  - There is another way of declaring CHARACTER length when PARAMETER attribute is used. The compiler assumed the length from the parameter and you do not have to state the length. The symbol * is used for the length of character.
    EXAMPLE ::
    CHARACTER (LEN = *), PARAMETER :: dir = “my program”

Features of Fortran Program 8/8

Variables Attributes

Various Fortran attributes may be specified for variables in their type-declaration statement.

- The PARAMETER attribute dictates that a constant is being defined. A variable declared with this attribute may not have its value changed within the program unit.
- The PARAMETER attribute is used after the data type keyword, followed by variable name followed by a = and then value for the variable.
  EXAMPLE :
  REAL, PARAMETER :: PI = 3.143
  INTEGER, PARAMETER :: TOTAL = 10

Features of Fortran Program 9/8

Variable Initialization

- A variable receives its value with
  - Initialization: It is done once before the program runs.
  - Assignment: It is done when the program executes an assignment statement.
  - Input: It is done with a READ statement.
- The variable can be initialized in their type-declaration statement
  EXAMPLE:
  REAL :: x = 0.05
  INTEGER :: gravity = 10
- It can also be initialized before execution statement.

Features of Fortran Program 10/8

Standard input/output (I/O) statement

- Fortran uses the READ * as standard input statement to read data into variable from the keyboard.
- Read statement is of the form: READ *, input_list
- If READ * has n variables in its list, there must be n Fortran declared constants.
- Each entered constant must have the type of the corresponding variable.
- Data items are separated by spaces and may spread into multiple lines.
- The asterisk * can be change to specify the data format. The format is of the form READ "(input_format)", input_list

Features of Fortran Program 11/8

Standard input/output (I/O) statement cont

- Fortran uses the PRINT * as standard output statement to display data output on the screen.
- Print statement is of the form: PRINT *, output_list
- To print undeclared string in your program put the string statement in between double quotes. e.g PRINT*,"This is fortran class" displays This is fortran class
- PRINT*, without output_list produces a blank line
- A formatted PRINT statement is of the form: PRINT "(output_format)", output_list

Features of Fortran Program 10/8

Program Comments

- Comments start with a exclamation mark (!)
- Everything following (!) will be ignored by the compiler
- Always comment your code so you can understand it later and also for others to understand when reading your code.

Example :

```
output = input1 + input2                    ! sum the inputs

PROGRAM first_program
! This program tries to illustrate some basic features of fortran

INTEGER :: x, y
```

Features of Fortran Program 11/8

Cases and Indentation

- Except within character strings, Fortran is completely case-insensitive.
- Fortran keywords such as REAL, DO, END, IF and intrinsic functions such as SIN, COS , TAN should be in “UPPER CASE“ letters and user created entities such as variable names should be in lower case.
- ’Indentation’, this is essential for program readability. It is common to indent a program’s content by 2 spaces from its header and END statement, and also indent the statements contained within, for example, Fortran keywords such as IF constructs or DO loops by a similar amount.

Operators in Fortran 1/6

- There are four types of operators in Fortran 90: arithmetic, relational, logical and character.

Arithmetic Operator

Order of Precedence	Operator	Meaning
[1]	**	Exponentiation operator
[2]	/	Division operator
[2]	*	Multiplicative operator
[3]	+	additive operator
[3]	-	Minus operator

- Within the same level of precedence, evaluation will proceed from left to right for all the operators with the exception of exponentiation where evaluation proceeds from right to left.
- Parentheses (...) can be used to change the default order of precedence.

Relational Operator

- Relational operators are used to test the relationship between two numeric operands for that specific operator.
- Expression involving these operators will reduce to either '.TRUE.' or '.FALSE.'.

Operator	Meaning
$<$ or .LT.	Less than
$>$ or .GT.	Greater than
$==$ or .EQ.	Identically equal to
$/=$ or .NE.	Not equal to
$<=$ or .LE.	Less than or equal to
$>=$ or .GE.	Greater than or equal to

Logical Operator

- Logical operators act on logical expressions to create a largest logical expression.
- Parentheses (...) can be used to change the default order of precedence.

Order	Operator	Meaning
[1]	.NOT.	Negative the following operand
[2]	.AND.	.TRUE. if both operands are .TRUE.
[3]	.OR.	.TRUE. if at least one operand is .TRUE.
[4]	.EQV.	.TRUE. if both operands reduce to the same
[4]	.NEQV.	.TRUE. if both operands are different

Character Operator

- There is only one character operator, concatenation symboled //.
- If strings A and B have lengths m and n, then concatenation A // B is a string of length m+n
- It is used to join two separate characters.

Example:
'Man' // 'chester' gives 'Manchester'

```
CHARACTER(LEN = 7) :: word_1 =“missing”
CHARACTER(LEN = 3) :: word_2 = “you”
CHARACTER(LEN = 10) :: ans
ans = word_1 // word_2          ! missingyou
```

Character substrings

- A consecutive portion of a string is a substring.
- To use substrings, one may add an extent specifier to the CHARACTER variable.
- An extent specifier has the following form:

  (integer_exp1 : integer_exp2)

- The first and the second expressions indicates the start and end: (3 : 8) means 3 to 8.
- If A = "abcdefg", then A (3 : 5) means A's substring from position 3 to position 5 (i.e., "cde")
- If the first extent specifier is missing, the substring start from the first character, and if the second extent specifier is missing, the substring ends at the last character.

Fortran Intrinsic Function 6/6

Function	Meaning	Data type
SQRT(x)	square root of x	Real
COS(x)	cosine of x	Real in radian
SIN(x)	Sine of x	Real in radian
TAN(x)	Tangent of x	Real in radian
ASIN(x)	Sine inverse	Real
ACOS(x)	Cosine inverse	Real
ATAN(x)	Tangent inverse	Real
EXP(x)	e^x	Real
ALOG(x)	$log_e(x)$	Real
ALOG10(x)	$log_{10}(x)$	Real
ABS(x)	Absolute value of x	Real, Integer
MAX(x1,x2 ....)	Maximum value of x vector	Real, integer
MIN(x1,x2 ....)	Minimum value of x vector	Real, integer
MODULO(x, y)	x modulo y	Real, integer
MOD(x, y)	remainder when x is divided by y	Real

Data type Conversions 6/6

Function	Meaning
INT(x)	Truncate a real number to an integer
NINT(x)	Round a real number to the nearest integer
REAL(x)	Convert to real
CEILING(x)	Nearest integer greater than or equal to x
FLOOR(x)	Nearest integer less than or equal to x
CMPLX(x)	Real to complex
FRACTION(x)	Fractional part of real

Formatted Output 1/7

- Lots of I/O related features in Fortran - not all will be considered for now
- Fortran PRINT statements do formatting by preceding the variables to be printed with (format descriptor).
- Format descriptor are put in parentheses.

Symbols used with format descriptor

Symbol	meaning
c	Column number
d	Number of digits to right of the decimal places of real
m	Minimum number of digits to be displayed
n	Number of spaces to skip
r	Repeat count - the number of times to use a descriptor
w	Field width - the number of characters to use

They are applied to both output and input

Integer Output - The I Descriptor

Integer values are right justified in their fields. This means that integers are printed out so that the last digit of the integer occupies the rightmost column of the field.
The I descriptor has the general form

rIw or rIw.m

where r, w, and m have the meanings given in the above table.

Example

```
INTEGER :: x = 12345
PRINT'(I5)', x          !print integer in field width of 5
```

If the integer is too large to fit into the field in which it is to be printed, then the field is filled with asterisk.

Real Output - The F Descriptor

Real values are printed right justified within the fields. If the specified field width is less than the number to be printed, the number is rounded off before it is displayed. On the other hand, if the specified width is more than the number to printed, extra zeros are appended to the right of the decimal point. The I descriptor has the general form

rFw.d

where r, w, and d have the meanings given in the table.

Example:

```
REAL :: x = 1234678
PRINT'(F8.3)', x     !print real of field width 8 with 3d.p
```

This will produce 1234.678

Real Output - The E Descriptor

Real data can be printed in exponential form using the E descriptor. Real values displayed by E descriptor are normalized to a range between 0.1 and 1.0. For example 4.096 x 10^3 will appears as 0.4096 E+04 using the E descriptor

The E descriptor has the general form

rEw.d

The width d of E descriptor must satisfied the expression $w \geq d + 7$ where d has the same meaning given in the table

Example:

```
REAL :: w = 23456
PRINT'(E11.5)', w
```

This produce 0.23456E+05 as output.

True Scientific Notation - The ES Descriptor

The ES descriptor is exactly the same as the E descriptor, except that the number to be printed is displayed with mantissa in the range between 1 and 10

The ES has the form:

rESw.d

The formula for the minimum width of an ES descriptor is the same as that of E but the leading zero in E descriptor is replaced by a significant digit in ES.

Example:

```
REAL :: w = 23456
PRINT'(ES11.5)', w
```

This produce 2.34560E+04 as output.

Character Output - The A Descriptor

A descriptor has the form:

rA or rAw

where r and w have the meanings given in the table

The rA descriptor displays character data with the same width as the number of characters being displayed whiles the rAw displays character data with width w. The rA are printed right justified and if the width of the field is shorter than the length of character variables only first w characters of the variable are printed.

Example:

```
CHARACTER(LEN = 7) :: dept = 'physics'
PRINT'(A7)', dept
```

This produce physics as output.

```
PRINT'(A3)', dept
```

This produce phy as output.

Horizontal Spacing - The X Descriptor

The X Descriptor is used to add one or more blanks space between two values on the output line.
The X Descriptor has the form:

nX

where n is the number of blanks to insert.
Example:

```
CHARACTER(LEN = 7) :: dept = 'physics'
PRINT'(A7)', dept
```

This produce physics as output.

```
PRINT'(2x,A7)', dept
```

This produce physics as output (ie moves two space to the right)

Recap of edit descriptors

Descriptor	Description	Example
I	Used for integer output. Takes the form rIw.m.	print "(3i5)", i, j, k
F	Used for real number output. Takes the form rFw.d.	print "(f12.3)",pi
E	Used for real output in exponential notation. Takes the form rEw.d. Width field: $w \geq d + 7$.	print "(e10.3)",123456.0 gives 0.123e+06
ES	Used for real output (scientific notation). Takes the form rESw.d. Width field: $w \geq d + 7$.	print "(es10.3)",123456.0 gives 1.235e+05

Edit descriptors conti.

Descriptor	Description	Example
A	Used for character output. Takes the form rAw.	print "(a10)", str.
X	Used for space output. Takes the form nX where n is the number of desired spaces.	print "(5x, a10)", str.
/	The Slash descriptor is used to insert blank lines. This takes the form / and forces the next data output to be on a new line.	print "(/,5x, a10)", str.

Formart descriptor symbols.

Descriptor	Description
c	Column number
d	Number of digits to right of the decimal place for real input or output
m	Minimum number of digits to be displayed
n	Number of spaces to skip
r	Repeat count. That is the number of times to use a descriptor or group of descriptors
w	Field width. That is the number of characters to use for the input or output

Descriptors conti

- Integer values are right justified in their fields. If the field width is not large enough to accommodate an integer then the field is filled with asterisks.
- Real values are right justified in their fields. If the field width is not large enough to accommodate the real number then the field is filled with asterisks.
- The real output width field must satisfy the expression: $w \geq d + 7$
- The Scientific notation has the mantissa ranging 1.0 to 10.0 unlike the E descriptor (mantissa range 0.1 to 1.0).
- If the character field width is not large enough to accommodate the character string then the field is filled with the first w characters of the string

Fortran - Basic Input Output

- So far we can read data from keyboard using the read(* *) statement, display output to the screen using the print * statement and write output to a file using the write(* *) statement. This form of input-output is free format I/O, and it is called list-directed input-output.
- The free format simple I/O has the form below:
  1. read(*,*) item1, item2, item3...
  2. print *, item1, item2, item3
  3. write(*,*) item1, item2, item3...
- The list-directed I/O statements are easy to use but we have no control over the appearance of the input and output. To have control over the I/O appearance, we use formats. Fortran formats are used to control the appearance of the input and output

Formatted Input Output

- The formatted input output has the syntax as follows:
  1. read *format specification, variable/output_list*
  2. print *format specification, variable/output_list*
  3. write *format specification, variable/output_list*
- Variable/output_list is a list of the variables to be read from keyboard/file, printed to screen or written to screen/file.
- Format specification defines the way in which formatted data is displayed. That is, it specifies the format at which the values in the output list are to be displayed. It consists of a string, containing a list of edit descriptors in parentheses.
- An edit descriptor specifies the exact format, for example, width, digits after decimal point etc., in which characters and numbers are displayed.

The format statement

The format statement allows you to repeat edit descriptors, mix and match character, integer and real output in one statement. For most of the cases, edit descriptors are separated by commas.

Format statement Example

```
program formatprint
implicit none
character (len=15) :: name
integer :: id
real :: weight
name = 'Atakpa'
id = 2586
weight = 73.2
print *,' The details of the student are'
print 88
88 format (7x,'Name:', 7x, 'Id:', 5x, 'Weight:')
print 200, name, id, weight
200 format (1x, a, 2x, i5, 2x, f5.2)
end program formatprint
```

The format statement conti.

You can also define the format and call in.

Format statement Example

```
program formatprint
implicit none
character (len=30) :: name, Format
integer :: id
real :: weight
name = 'Atakpa'
id = 2586
weight = 73.2
Format = "(5x, a, 2x, i3, 2x, f5.2)"
WRITE(*,Format) name, id, weight
end program formatprint
```

Format Print Example

Example

```
program formatprint
implicit none
real :: c = 1.2786456e-9, d = 0.1234567e3
integer :: n = 459789, k = 78, i = 9
character (len=15) :: str="Tutorials Point"
print "(i6)", k
print "(i6.3)", k
print "(3i10)", n, k, i
print "(i10,i3,i5)", n, k, i
print "(a15)",str
print "(f12.3)", d
print "(e12.4)", c
print '(/,3x,"n = ",i6, 3x, "d = ",f7.4)', n, d
end program formatprint
```

Fortran 90 Control Structures 1/5

Three block constructs

1. IF Statement and IF Constructs
2. DO
3. CASE

All can be nested.

IF statement and Construct

An IF Statement/Construct allows part of a program to be executed if certain condition is true.
Relational and logical operators are used to specify then condition
Fortran 90 has IF statement or logical IF and IF construct

- IF Statement or Logical IF
- IF Constructs
  Three form of IF constructs
  - IF-THEN-END IF
  - IF-THEN-ELSE-END IF
  - IF-THEN-ELSE IF-END IF

Logical IF or IF Statement

The logical IF or IF Statement executes or does not execute a statement based on the value of a logical expression.
The syntax of the logical IF statement is:

IF (<logical-expression>) < action-statement>

The action-statement is executed if logical-expression is true. This is used if there is only one action statement. It can also be used to exit from a loop.

Example

```
IF (mark > 70) PRINT*, 'A'
```

IF Construct - (IF-THEN-END IF)

This form simply performs a single test and if the test result is true then a series (or a single) Fortran statements that lie in the 'body' of the construct are executed.
The syntax is of the form:

```
IF (< logical-expression >) THEN
  < action-statement1 >
  < action-statement2 >
  <        etc        >
END IF
```

Remember to do indentation to make your code readable !!!

IF Construct - (IF-THEN-ELSE-END IF)

This form of IF Construct allows the program to take action on a series of Fortran statement if the condition is true as well as if it is false. This is done by the addition of the 'ELSE' keyword.
The syntax is as follows:

```
IF (< logical-expression >) THEN
   < Statements if condition is true >
ELSE
   < Statements if condition is false >
END IF
```

Remember to do indentation to make your code readable !!!

IF-THEN-ELSE-END IF Example

```
PROGRAM if_prog1                          ! start of program
  IMPLICIT NONE                           ! all variables must be declared
  INTEGER :: x, y                         ! x, y declared as integers
  PRINT*, 'Please enter two integers'     ! print prompt
  READ*, x, y                             ! read in x and y
  IF (x < y) THEN
    PRINT*, 'The first number is the smaller.'
    x = y - x
    PRINT*, 'The difference is ', x
  ELSE
    x = y - x
    PRINT*, ' The difference is negative', x
  END IF
END PROGRAM if_prog1
```

IF Construct - (IF-THEN-ELSE IF-END IF)

This form of IF Construct allows the programmer to produce a 'mutually exclusive' list of options. Note there can be any number of 'ELSE IF' branches in the construct and also 'ELSE' is optional.
The syntax is as follows:

```
IF (< logical-expression1 >) THEN
  < Statements if condition1 is true >
ELSE IF (< logical-expression2 >) THEN
  < Statements if condition2 is true >
ELSE IF (< logical-expression3 >) THEN
  < Statements if condition3 is true >
ELSE
  < Statements all conditions are false >
END IF
```

DO and DO WHILE loops

If a block of code is to be performed repeatedly it is put inside a DO loop
There are two basic types of DO loops:

- Deterministic DO loops
  The number of times the section is repeated is stated explicitly.
- Non-deterministic DO loops
  The number of repetitions is not stated in advance.

DO loops - (Deterministic DO loop)

The number of times the section is repeated is stated explicitly. The syntax is as follows:

```
DO control-var = initial, final [ , step ]
  < Statements >
END DO
```

- control-var is an INTEGER variable, initial, final and step are INTEGER expression; however, step cannot be zero.
- If step is omitted, its default value is 1
- statement are executable statement of the DO
- The control-var runs from initial in increments of step until it reaches the value final and then the loop terminates.
- If step is positive, this DO counts up; if step is negative, this DO counts down

DO loops - (Deterministic DO loop)

If step is positive

- The control-var receives the value of initial
- If the value of the control-var is less than or equal to the value of final, the statements part is executed. Then, the value of step is added to control-var, and goes back and compares the value of control-var and final.
- If the value of control-var is greater than the value of final, the DO-loop completes and the statement following END DO is executed.

DO loops - (Deterministic DO loop)

If step is negative

- The control-var receives the value of initial
- If the value of control-var is greater than or equal to the value of final, the statements part is executed. Then, the value of step is added to control-var, goes back and compares the value of control-var and final.
- If the value of control-var is less than the value of final, the DO-loop completes and the statement following END DO is executed.

DO loops - (Non-deterministic DO loop)

The number of repetitions is not stated in advance. The enclosed section is repeated until some condition is or is not met. Note that this loop can be infinite without the condition.
This may be done in two alternative ways:

- The first requires a logical reason for stopping looping
- The second requires a logical reason for cycling looping.

Non-deterministic DO-loops are particularly good for:

1. summing power series (looping stops when the absolute value of a term is less than some given tolerance)
2. single-point iteration (looping stops when the change is less than a given tolerance)

Non-deterministic Do loop - (with Exit condition)

In these, the loop is terminated by a conditional exit statement, ie. an IF statement which executes exit action.
The syntax is as follows:

```
DO
  < statement body >
  IF (< logical-expression > ) EXIT
  < statement body >
END DO
```

The loop continues until some logical expression evaluates as TRUE. Then it jumps out of the loop and continues with the code after the loop. In this form a TRUE result tells you when to stop looping.

Non-deterministic Do loop - (with Cycle condition)

The CYCLE statement starts the next iteration (ie. executing statements again).
The syntax is as follows:

```
DO
  < statement body >
  IF (< logical-expression > ) CYCLE
  < statement body >
END DO
```

The action of the IF statement if it is TRUE is to cause the program to cycle immediately to the next iteration of the loop, skipping any remaining statements in the current cycle.

Non-deterministic Do loop - (Exit and Cycle Example)

```
PROGRAM exit_cycle_exam
  IMPLICIT NONE
  INTEGER :: i
  INTEGER, PARAMETER :: istart = 1, iend = 5, ilimit=3
  DO i = istart, iend
    IF (i == ilimit) EXIT
    PRINT*, i
  END DO
END PROGRAM exit_cycle_exam
```

Please re-run the program by replacing the EXIT with CYCLE and exam the difference in the output.

Nested DO Loops

If one loop is completely inside another one, the two loops are called nested loops.
For each control-var of the outer loop, the entire inner loop has to be completed before next iteration in the outer loop.
When Fortran compiler encounters an END DO statement, it associates that with the innermost currently open loop. Therefore, the first END DO statement closes the inner loop and the last the outermost loop.

Nested DO Loops -Example

```
PROGRAM nest_exam
 IMPLICIT NONE
 INTEGER :: x, y
  INTEGER, PARAMETER :: xstart = 1, xend = 2
  INTEGER, PARAMETER :: ystart=1, yend=4
  REAL :: results
  PRINT*,' x y results'
  DO x = xstart, xend
   DO y = ystart, yend
    result = (x*1.0)/(y*1.0)
    PRINT*, x, y, result
   END DO
  END DO
PROGRAM nest_exam
```

DO WHILE construct

The DO WHILE loop always exits at the top of the loop construct as this is where the test takes place.
The syntax is as follows:

```
DO WHILE (< logical-expression >)
  < body statements >
END DO
```

The body statements in the DO WHILE block are executed for as long as the result of the logical-expression remains TRUE.

SELECT CASE Construct

The SELECT CASE statement chooses which operations to perform, depending on the set of outcomes (selector) of an expression than can be integer, character or logical, but not real.
The general form is:

```
SELECT CASE (expression)
  CASE (case_selector_1)
    < statements_1 >
  CASE (case_selector_2)
    < statements_2 >
  CASE DEFAULT
    < statements_default >
END SELECT
```

SELECT CASE Construct

If the expression is in the range of values included in case_selector_1, then < statements_1 > is executed and then move to the next case_selector to execute statements in that block. The CASE DEFAULT (optional)is used whenever the value expression is outside of all the case selector.
The SELECT CASE construct is alternative to an IF-ELSE IF - END IF construct.

Lecture Outline

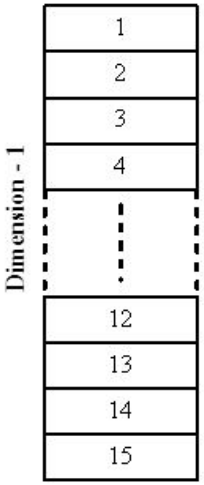
- Introduction to Arrays
- Declaring Arrays
- Assigning Values
- Array Related Terms
- Array Sections
- Multi-dimensional arrays

Introduction to arrays

- In fortran, arrays are a collection of variables of the same type in columns and rows.
- All arrays have memory locations. The lowest address corresponds to the first element and the highest address to the last element.
- Arrays can be:
  1. One dimensional (like vectors)
  2. Two-dimensional (like matrices), etc
- Fortran allows you to create up to 7-dimensional arrays.

Array Example

Arrays



One dimension array

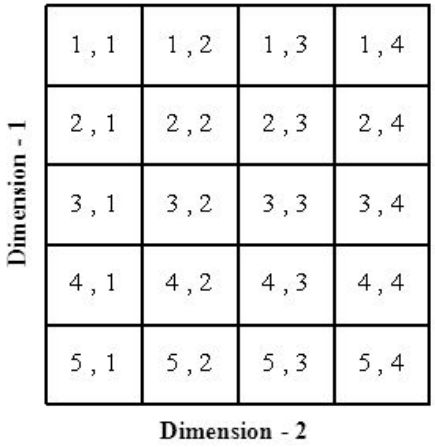
Vector(i)

Rank: 1

Extent: 15

Shape: Rank, extent

Size: 15



Two dimension array

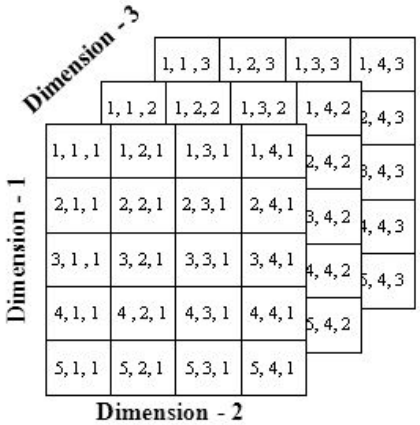
Vector(i , j)

Rank: 2

Extent: 5 , 4

Shape: Rank, extent

Size: 20



Three dimension array

Vector(i, j, k)

Rank: 3

Extent: 5, 4, 3

Shape: Rank, extent

Size: 60

Some Array Related Terms

Term	Meaning
Rank	It is the number of dimensions an array has. eg, integer :: tx(3,3), tx has a 2 ranks
Extent	It is the number of elements along a dimension. eg integer :: tx(3,3), tx has extent 3 in both dimensions. integer :: tx(10) has extent 10.
Shape	The shape of an array is characterized by its rank and the extents of each of its dimensions. eg for the array tx, the shape is (3, 3).
Size	It is the number of elements an array contains. eg The array integer :: tx(3,3) has a size of 9

Declaring Arrays

- Array declarations in Fortran go at the beginning of the program, before any executable statement.
- Arrays can be declared with either a DIMENSION attribute or with a type declaration. The type declarationis mostly preferred, because it is best anyway to declare the type of the array.
- The DIMENSION attribute requires three components in order to complete an array specification. They include: rank, shape, and extent

Example

One dimensional array x of real numbers containing 100 elements
REAL, DIMENSION(100) :: x
A two-dimensional array yy of integers containing 25 elements:
INTEGER, DIMENSION (5,5) :: yy
A three dimensional array mm of real containing 60 elements
REAL, DIMENSION(5,4,3) :: mm

Declaring Arrays continues

- The individual elements of arrays are referenced by specifying their subscripts. The first element has a subscript of one, the second two, third three etc.

more Example

- INTEGER, DIMENSION(3:8) :: n: one dimensional array n of integers, indexed from 3 to 8.
- DOUBLE PRECISION, DIMENSION(4,5) :: c: two-dimensional array c of double precision real numbers, the first index running from 1 to 4, and the second from 1 to 5.
- CHARACTER, DIMENSION(30)*20 :: student: one-dimensional array student of strings, indexed from 1 to 30, each string up to 20 symbols long.
- REAL, DIMENSION(0:5,1:10,-3:3) :: num: three-dimensional array num of single precision real numbers, the first index running from 0 to 5, the second from 1 to 10, and the third from -3 to 3.

Array declaration

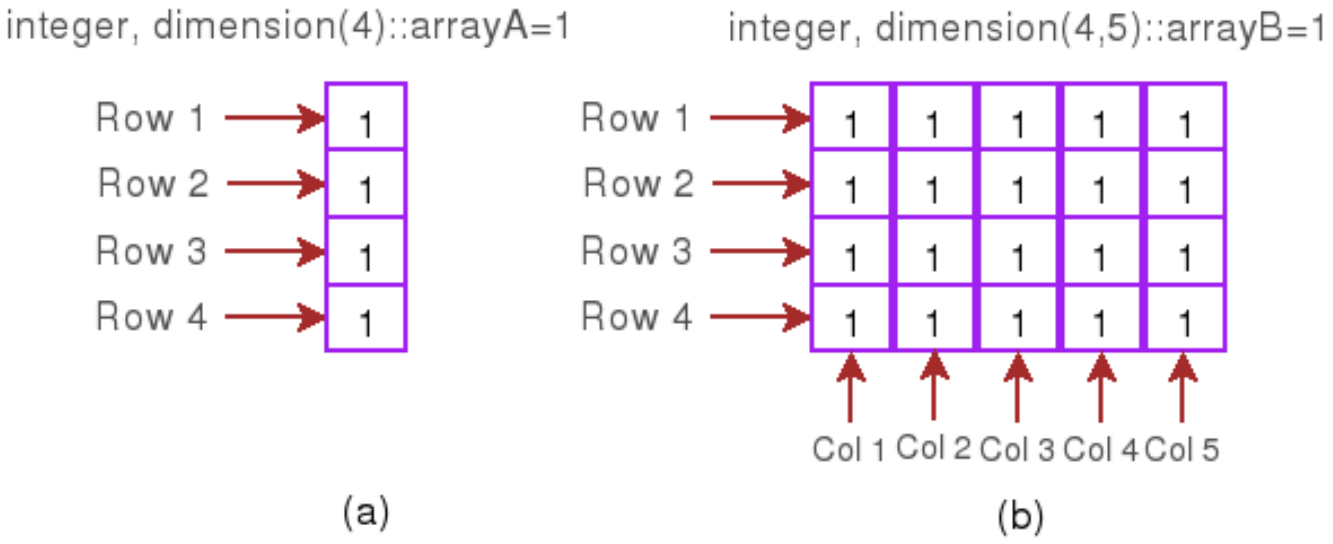


Figure 1: One and Two dimensional arrays

Example code

Average of 10 numbers

```
PROGRAM average
IMPLICIT NONE
REAL ,DIMENSION(10) :: x
REAL :: avg,adn
INTEGER :: i
PRINT *, 'enter 10 numbers'
adn=0.0
DO i=1,10
read *, x(i)
adn=adn+x(i)
END DO
average=adn/10
PRINT *, 'the average is ',avg
END PROGRAM average
```

Assigning Values

- You can either assign values to individual members or use a loop.

Example

```
REAL, DIMENSION(6) :: a
! Assign individual members example
a(1) = 2.0
a(2) = 8.0
a(3) = 5.5, etc
! Using a loop example
do i=1,6
a(i) = i * 2.0
end do
```

- One dimensional array elements can be directly assigned values using a short hand symbol, called array constructor:
  a = (/0.5, 2.3,4.5,0.9,7.2,10.0/)

Multi-dimesional array example

By multi-dimensional array we mean more than one dimension.

example

```
PROGRAM colrow
IMPLICIT NONE
INTEGER, DIMENSION(5,5) :: a
INTEGER:: i,j,sum
DO i = 1, 5
sum=0
DO j = 1, 5
sum=sum+1
a(i,j)=sum
END DO
END DO
DO i=1,5
DO j =1,5
PRINT *,a(i,j)
END DO
END DO
END DO colrow
```

Multi-dimensional arrays: two dimensional arrys

Example

```
PROGRAM multi
IMPLICIT NONE
INTEGER :: tx(3,3), i , j ! two dimensional array
DO i=1,3
DO j = 1, 3
tx(i, j) = i+j
END DO
END DO
OPEN(10,file="data.txt",status='new') DO i=1,3
DO j = 1, 3
WRITE(10,'(I3,2X,I3,2X,F5.2)')i,j,tx(i,j)
END DO
END DO
END PROGRAM multi
```

Array Sections

- Fortran provides an easy way to refer to several elements, or a section of an array, using a single statement.
- To access an array section, you need to provide the lower and the upper bound of the section, as well as a stride (increment), for all the dimensions.

Array section syntax

array ([lower]:[upper][:stride], ...)

- When no lower and upper bounds are mentioned, it defaults to the extents you declared, and stride value defaults to 1.

Array section example

example

```
PROGRAM arraysection
IMPLICIT NONE
REAL,DIMENSION(10) :: a, b
INTEGER:: i, asize, bsize
a(1:7) = 5.0 ! a(1) to a(7) assigned 5.0
a(8:) = 0.0 ! rest are 0.0
b(2:10:2) = 3.9
b(1:9:2) = 2.5
!display
asize = size(a)
bsize = size(b)
DO i = 1, asize
PRINT *, a(i)
END DO
DO i = 1, bsize
PRINT *, b(i)
END DO
END PROGRAM arraysection
```

Sample quiz on arrays

Write SHORT NOTES WITH EXAMPLES on the following:

1. A fortran array
2. Declaring an array in fortran
3. Referencing an array in fortran
4. Assigning values to an array in fortran
5. Refering to a section of an array in fortran

I/O functionalities: introduction

- In the last section, we learned how to read data from, and write data to the terminal.
- In this section we will study file input and output functionalities provided by Fortran.
- Fortran allows you to read data from, and write data into files.
- You can read and write to one or more files using the OPEN, READ, WRITE and CLOSE statements.
- Before using a file you must open the file.
- Once done using the file you must close it.

Opening and Closing Files

- The open command is used to open files for reading or writing.
- The simplest form of the command is:
  open (unit = number, file = "name", status = "sta").
- The open statement may have a general form:
  open (list-of-specifiers)
- After the file has been opened, it is accessed by read or write statements.
- Once done, it should be closed using the close statement.
- The close statement has the following syntax:
  close ([UNIT=]u[,IOSTAT=ios,ERR=err,STATUS=sta])

Reading from and Writing into Files

- The read and write statements respectively are used for reading from and writing into a file respectively.
- To write to a file you need the OPEN, WRITE and CLOSE statements.
- To read from a file you need the OPEN, READ and CLOSE statements.
- They have the following general syntax:
  read ([UNIT=]u, [FMT=]fmt, IOSTAT=ios, ERR=err, END=s)
  write([UNIT=]u, [FMT=]fmt, IOSTAT=ios, ERR=err, END=s)
- The END=s specifier is a statement label where the program jumps, when it reaches end-of-file.
- The list of specifiers are discussed in the next slides

Commonly used list-of-specifiers:

Specifier	Description
[UNIT=] u	The unit number, u, could be any number in the range 9-99 and it indicates the file. Every open file in the program must have a unique number.
IOSTAT= ios	It is the I/O status identifier and should be an integer variable. If the open statement is successful then the ios value returned is zero else a non-zero value.
ERR = err	It is a label to which the control jumps in case of any error.
FILE = fname	File name, a character string.

specifiers conti

Specifier	Description
STATUS = sta	It shows the prior status of the file. Status is a character string and can have one of the ff three values: NEW, OLD or SCRATCH.
ACCESS = acc	It is the file access mode. Can have either of the two values:SEQUENTIAL or DIRECT. The default is SEQUENTIAL.
FORM= frm	It gives the formatting status of the file. Can have either of the two values: FORMATTED or UNFORMATTED. The default is UNFORMATTED.
RECL = rl	It specifies the length of each record in a direct access file.

Open close example

This example demonstrates opening a new file for writing some data into it.

example

```
program openclose
implicit none
real, dimension(100) :: x, y
real, dimension(100) :: p, q
integer :: i
do i=1,100
x(i) = i * 0.1
y(i) = sin(x(i)) * (1-cos(x(i)/3.0))
end do
open(1, file='data1.dat', status='new')
do i=1,100
write(1,*) x(i), y(i)
end do
close(1)
end program openclose
```

Format write Example

Assuming we want to write the formated output into a file.

write example

```
program formatwrite
implicit none
real :: c = 1.2786456e-9, d = 0.1234567e3
integer :: n = 459789, k = 78, i = 9
character (len=15) :: str="Tutorials Point"
open(unit=10, file="test.txt")
write(10,'(i10,2x,i4,2x,i5,2x f12.3,2x,e12.4,2x,a15 )') n,k,i,d,c str
close(10)
end program formatwrite
```

Read from and writing to file example

```
PROGRAM readwrite
IMPLICIT NONE
!
INTEGER,PARAMETER:: il=75
REAL, DIMENSION(100) :: p,q
INTEGER :: ic,k
! opening the file for reading
OPEN(99,file='temp.txt', status='old')
ic=0
DO k=1,il
READ (99,*,end=100) p(k),T(k)
ic=ic+1
END DO
100 continue
CLOSE (99)
You can then loop over ic and write to an output
END PROGRAMM readwrite
```

Quiz 2

Given a file Offinso.txt with the format:
1983 01 01 0.00
1983 01 02 0.00
1983 01 03 0.00
1983 01 04 -9.00
1983 01 05 -9.00
.....
.....
2016 12 30 0.00
2016 12 31 0.00
Open this file, read it and write the output into another file call Atakpa.txt

Fortran Program Units

- A fortran program can be built up from a collection of program units
- There are four types of program unit in the Fortran
  1. Main program
  2. Functions
  3. Subroutines
  4. Modules
- Each program must contain one (and only one) main program
- A subprogram or procedure (functions and subroutines) are computations that can be "called" (invoked) from the main program
- Relevant functions and subroutines can be put together into a module.

Fortran Subprogram 1

- Fortran 90 has two types of subprograms, functions and subroutines.
- Subprograms are independent blocks of Fortran which perform a specific task.
- They may be written and compiled separately, and are joined to a main program when the program is linked.
- One main advantage of subprograms is that they may be written and tested as isolated units to ensure that they operate correctly.
- Another advantage of subprograms is that they are portable, i.e it can be use in later programs whenever you need it.
- Functions and subroutines may be internal (i.e. contained within and only accessible to one particular program unit) or external (and accessible to all).

Fortran Subprogram 2

Functions syntax 1

- Function returns a single computed result via the function name.
- A Fortran function, or function subprogram has the following syntax:

```
type FUNCTION function-name (arg1, arg2, . . ., argn)
   IMPLICIT NONE
   [ declaration part ]
   [ execution part ]
END FUNCTION function-name
```

- type is a Fortran 90 type (e.g., INTEGER, REAL, LOGICAL, etc).
- function-name is a Fortran 90 identifier.
- arg1, ..., argn are dummy arguments

Functions syntax 2

- Function receives inputs from the outside world via its dummy arguments, does some computations, and returns the result with the function-name
- The function-name must appear in one or more assignment statement like this:

  function-name = expression

  where the result of expression is saved to the name of the function
- Note that function-name cannot appear in the right-hand side of any expression.
- The variables listed in the argument list in the function definition are called dummy variables. Their names do not have match the names used in parts of the program where the function is called.

Functions syntax 3

Note that a function without dummy arguments is declared with empty parentheses ( ) e.g

```
type FUNCTION function-name ( )
   IMPLICIT NONE
   [ declaration part ]
   [ execution part ]
END FUNCTION function-name
```

- If the function is written in the same file as the main program, it must appear after the end of the main program.
- The arguments used when the function is actually invoked are called actual arguments. They may be variables (e.g. x, y), constants (e.g. 1.0, 2.0) or expressions (e.g. 3.0 + x, 2.0/y), but they must be of the same type and number as the dummy arguments.

Recursive Functions syntax 3

- A function that calls itself (directly or indirectly) is called a recursive function.
- If the function is recursive then the FUNCTION keyword in the top line must be proceeded by the keyword RECURSIVE.
- Its declaration is of the form:

```
type RECURSIVE FUNCTION function-name (arg1, . . . , argn)
  IMPLICIT NONE
  [ declaration part ]
  [ execution part ]
END FUNCTION function-name
```

Functions Intent Declaration 4

- It is a good practice to declare whether dummy arguments are intended as input or output using the INTENT attribute.
- The intent of a variable in a function/subprogram can have one of three values: in, out or inout
  INTENT(IN) - The dummy argument is used only to pass input data to the subprogram and its content does not change.
  INTENT(OUT) - The dummy argument is a variable and is used only to return results to the main program when the subprogram is invoked.
  INTENT(INOUT) - The dummy argument is used both to pass input data to the subprogram and also to return results to the main program when he subprogram is invoked.
- Us the intent attribute to help make sure that variables cannot be changed accidentally in a function.

Subroutine syntax 1

Subroutine is used in computations that involves the return of several values of different types. It takes values from its formal arguments, and returns some computed results with its formal arguments.
Subroutine syntax is of the form:

```
SUBROUTINE subroutine-name (arg1,arg2, ..., argn)
  IMPLICIT NONE
  [ declaration part ]
  [ execution part ]
END SUBROUTINE subroutine-name
```

If a subroutine does not require any dummy arguments, the first line is written with or without empty parentheses ( ) but the keyword subroutine and its name should be there.

To call Subroutine 2

- The subroutine is invoked by the CALL statement in the main program
- The CALL statement may have one of the three forms:
  1. CALL subroutine-name (arg1, arg2, ..., argn)
  2. item CALL subroutine-name ( )
  3. CALL subroutine-name
- The last two forms are equivalent and are used for calling subroutine without dummy arguments.
- The order and type of the actual arguments in the arguments list in the CALL statement must match the order and type of the dummy arguments declared in the subroutine.

Recursive Subroutine 3

- It is permissible for subroutine also to invoke itself.
- recursive subroutine must have the keyword recursive it its first line.
- Its declaration is of the form:

```
RECURSIVE SUBROUTINE subroutine-name (arg1,arg2, ..., argn)
 IMPLICIT NONE
 [ declaration part ]
 [ execution part ]
END SUBROUTINE subroutine-name
```

Fortran 90 Modules 1

- Modules exist so that anything required by more than one program unit may be package in a module and made available were needed.
- They are designed to hold declaration of variables, data and subprograms which are to be made available to other program unit.
- A program may use any number of modules, with the restriction that each must be named separately.
- A module looks like the main program, except that it does not have the executable part. Hence, a main program must be there to use modules
- A module consists of two part: a specification part for the declaration statements, and a subprogram part.

Fortran 90 Modules 2

Module Syntax

- Fortran 90 module has the following syntax:

```
MODULE module-name
  IMPLICIT NONE
  [ Declaration part ]
CONTAINS
  [ internal functions/subroutines ]
END MODULE module-name
```

- Modules can contain just the declaration part or the subprogram part, or both.
- Any program unit that makes use of the module's contents should include the following statement immediately after the program name or other program unit statement:

```
USE module-name
```

Fortran 90 Modules 3

Some features of Module

- After USE module-name statement, the variables and subprograms contained in the module-name can be used in the program unit.
- Note that it is not necessary to declare the module variables in the main program. The USE statement is sufficient.
- If a module is contained in the same file as the main program, it must come before the main program.
- Modules cannot USE themselves either directly (module A uses A) or indirectly (module A uses module B which uses module A)
- To restrict program unit to access only variables that it requires from module, the ONLY qualifier is used. e.g
  USE module-name, ONLY: var1, var2
  Only var1 and var2 of the module variables are accessible.

Fortran 90 Modules 4

Some features of Module

- The (USE module-name, ONLY) statement safeguards the module elements that are not needed by making them inaccessible to the program.
- It can also make programs more transparent, by showing the origin of the dtatobjects or subprograms, particularly if the program uses several modules.
- If the name of an item that is accessed from a module conflicts with another name elsewhere, one may use the "renaming" feature of USE.
- For each identifier in USE to be renamed, the the syntax:
  USE module-name, name-in-program => name-in-module
- In this program, the use of name-in-program is equivalent to the use of name-in-module in the module

Module accessibility - Public and Private

- By default, everything in a module is publicly available, that is, the USE statement in the main program makes available all of the code in the module
- PUBLIC and/or PRIVATE statements (or attributes) may be used to restrict access to some variables, declaration statement or subprograms in a module
- Everything that is declared PRIVATE is not available outside the module unit, PUBLIC is the opposite.
- As a statement PUBLIC or PRIVATE can set the default for the module, or can be applied to a list of variables or module names.
- As an attribute PUBLIC or PRIVATE can control access to the variables in a declaration list.

Module accessibility - Public and Private

- To specify PRIVATE and PUBLIC, do the following:
  PUBLIC :: name-1, name-2, ..., name-n
  PRIVATE :: name-1, name-2, ..., name-n
- The PRIVATE statement without a name makes all entities in a module private. To make some entities visible outside the module declare them as PUBLIC.
- PUBLIC and PRIVATE may also used in the declaration statement. e.g
  INTEGER, PRIVATE :: sum1, avg1
  where sum1 and avg1 are visible within the module
  INTEGER, PUBLIC :: sum2, avg2
  where sum2 and avg2 are available within/outside the module

Compiling Programs with Module

- The module may be in the same source file as other units or it may be in a different file.
- To enable compiler to operate correctly, the module must be compiled before any program units that USE it. Hence,
  1. if it is in a different file the module must be compiled first;
  2. if it is in the same file the module must come before any program units that USE it.
- Suppose a program consists of the main program mypro.f90 and a module mymod.f90, then compilation syntax is as follows:
  compiler_name -o mypro mymod.f90 mypro.f90
- Run the mypro executable file (./mypro.exe) to get the output

Compiling Programs with Module

- Since modules are supposed to be designed and developed separately, they can also be compiled separately to object codes.
- Compile the module with the syntax:
  compiler_name -c mymod.f90
- The -c option causes the file to be compiled but not linked; this creates mymod.o and mymod.mod files.
- Compile the main program with the syntax:
  compiler_name -c mypro.f90
- Once the two parts have compiled successfully, link them together with the command:
  compiler_name -o mypro mymod.f90 mypro.f90
- Run the mypro executable file (./mypro.exe) to get the output

Thank You !